NIE25143 – Big Data & AI/Machine Learning in Education

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Introduction

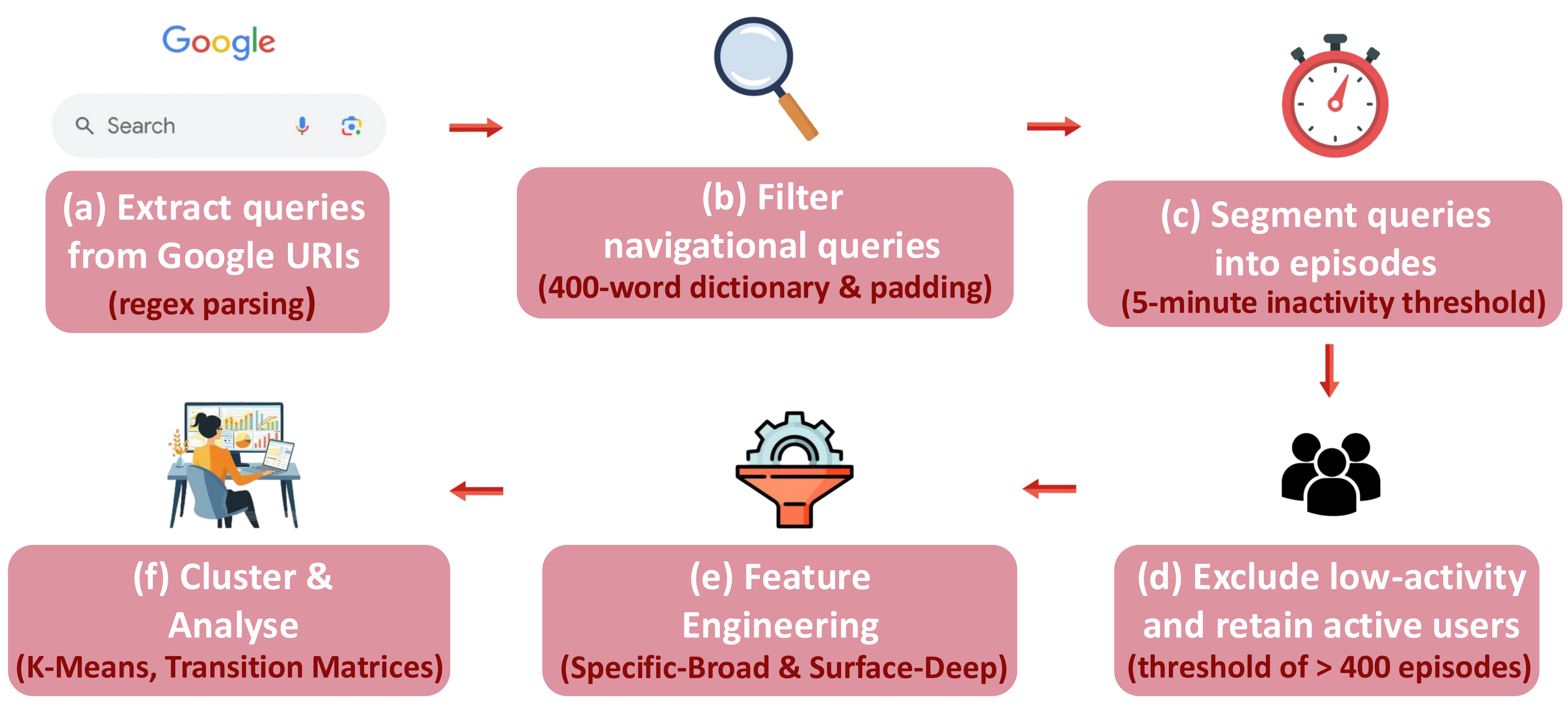
With Singapore's nationwide rollout of **Personal Learning Devices (PLDs)**, Google Search has become a routine tool for secondary students' self-directed learning. However, there is limited research on using **large-scale query logs** to understand how students formulate information needs and how these needs evolve over time.

Thus, this study examines the search behaviour of 4,961 students along two complementary dimensions:

- **Specific-Broad (S-B):** How granular is a query? Does the student zoom in on a specific entity, or stay broad and general?
- **Surface-Deep (S-D):** How cognitively engaged is the search? Is the student doing a quick lookup (surface) or iteratively refining toward understanding (deep)?

Methodology

1. Data Preprocessing Pipeline



The research was approved by NTU Institutional Review Board.

2. Feature Engineering

(a) S-B Dimension

Captures the **granularity** and **topical scope** of queries within an episode.

Features Engineered:
Query length, Lexical Diversity, Semantic Diversity, Entity Diversity, Topic Diversity, Hypernym Specificity, Brysbaert Concreteness Rating

(b) S-D Dimension

Captures **cognitive depth** and **intent complexity** across a search episode.

Features Engineered:
WH-Questions (who/what/why/when/how), Closure-Seeking Queries, Comparative Queries, Within-Episode Reformulation, Open/Closed and Possibility/Non-Possibility Type Questions

Results & Discussion

S-B Finding 1: Most Episodes Stay on One Topic

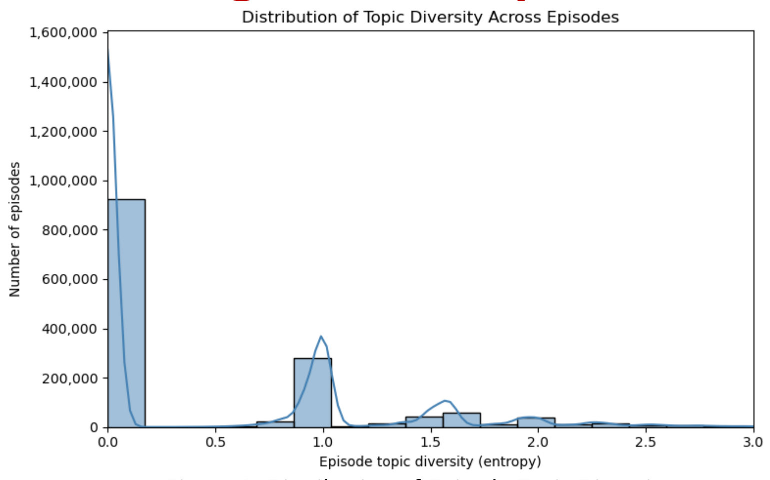


Figure 1: Distribution of Episode Topic Diversity

- Most episodes have **near-zero topic diversity** and stay on **one subject**.
- Only a **minority** of sessions span multiple topics.

Insight: Students tend to stay within **one subject per session** rather than broadly exploring multiple topics simultaneously.

S-B Finding 2: Students Do Not Search Linearly

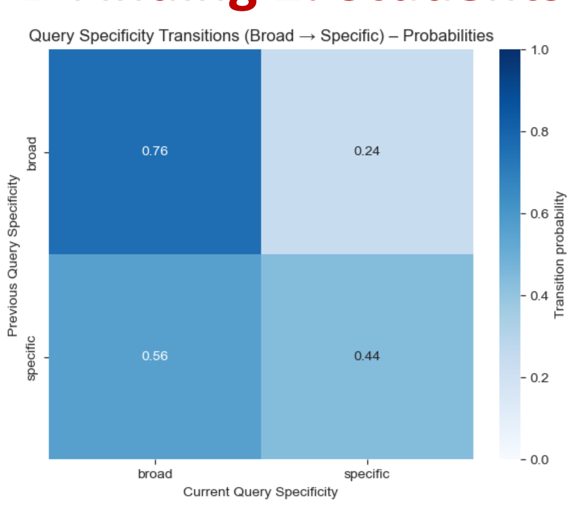


Figure 2: Query Specificity Transition Matrix

- After issuing a specific query, students **revert to broad search 56% of the time**, challenging the funnel-narrowing model.

Insight: Information-seeking is an **iterative, oscillating process** instead of progressive refinement.

S-D Finding 1: Most Searches are Shallow, Deep Thinking is Rare

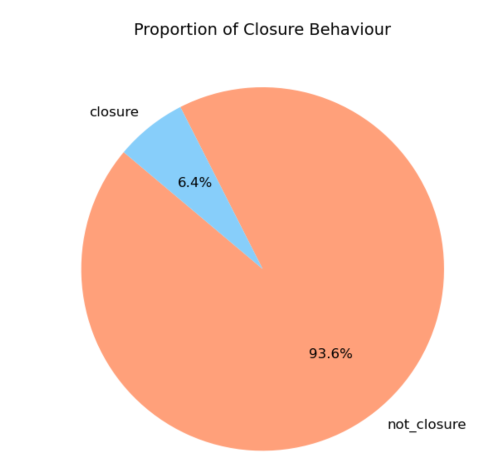


Figure 3: Percentage of Searches with Closure

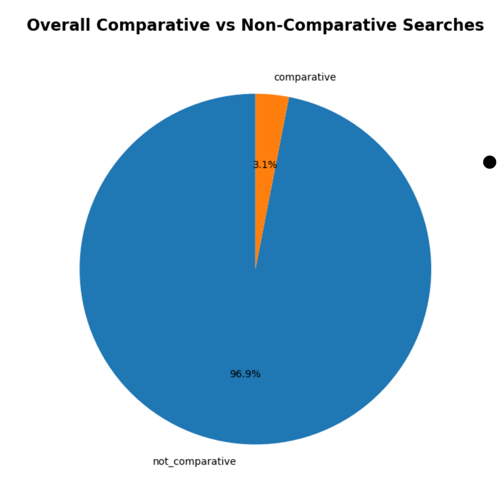


Figure 4: Percentage of Searches with Comparison

- Only **6.4%** of queries are closure-seeking and **3%** are comparative.

Insight: Most students phrase searches as **short keyword statements** instead of questions or evaluations. **Explicit intent** is **rare**.

S-D Finding 2: Longer Sessions Have More Refined Searches

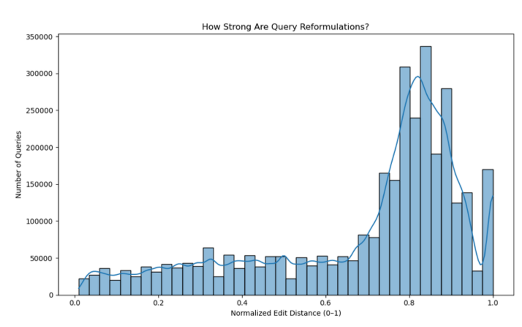


Figure 5: Distribution of Reformulation Strength at Query Level

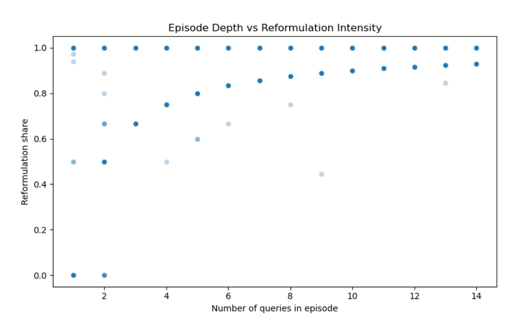


Figure 6: Reformulation Intensity as Episodes Get Deeper

- Most students cluster around normalised edit distance of 0.7-0.9, meaning consecutive queries **differ by most of their characters**.
- Deeper episodes show **near-complete query rewrites**, with reformulation share approaching 1.0.

Insight: Students' **extended search sessions** reflect **genuine iterative refinement**, not repeated identical lookups.

Conclusion

- **Students alternate between broad and specific queries**, reflecting exploration rather than systematic narrowing.
- **High search volume does not imply deep inquiry** as most queries are plain keywords with intent signals such as closure and comparative queries being rare.
- **Students exhibit active thinking during prolonged search sessions**, substantially rewriting their queries.

Future Work

- Validate search behaviour profiles **against outcomes** to assess educational impact.
- Address high school-level clustering (ICC=0.77) with a **3-level hierarchical model** and collect demographic data for subgroup analysis.
- Extend framework to **post-GenAI search behaviour** to benchmark against their pre-AI baseline and inform PLD monitoring dashboards.

References

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